



CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: Research Assistant Platform

Group No: 10, CSE471 Lab Section: 10, Summer 2026		
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Project Overview

The Research Assistant Platform is a web-based system that supports students and researchers across the full research lifecycle - from organizing and reading papers, to identifying research gaps, to managing collaborative research projects. The system supports four user roles: Researcher, Team Member, Supervisor, and Admin. It integrates meaningful AI tooling (summarization, insight extraction, gap analysis, an interactive paper-chat assistant, and title generation) rather than adding AI for its own sake, and gathers the heavier AI features behind a Free / Premium subscription model enforced through a monthly usage quota.

Tech Stack:

- **Language:** Python is used for both the frontend and backend layers.
- **Backend Framework:** FastAPI handles asynchronous REST API endpoints and data processing.
- **Frontend Framework:** Flask manages web page routing and renders Jinja2 templates.
- **Styling:** Tailwind CSS provides responsive, utility-first UI styling directly in HTML.
- **Database:** PostgreSQL stores relational data. It uses the pgvector extension for embeddings.
- **ORM:** SQLAlchemy manages database queries and object-relational mapping.
- **Background Processing:** Celery and Redis handle heavy PDF parsing, chunking, and embedding tasks.
- **Authentication:** JWT-based secure token management.
- **Deployment:** Hosted on Render or Railway (Backend) and Vercel or similar (Frontend).

External APIs:

- **Google Gemini API (gemini-2.0-flash):** Powers text summarization, insight extraction, gap analysis, question generation, and paper chat.
- **SSLCommerz / Stripe API:** Processes premium subscription transactions via secure payment gateways.
- **SMTP / Gmail API:** Dispatches automated email notifications for project milestones and system deadlines.

User Roles

1. **Researcher:** The primary user - uploads and organizes papers, takes notes, uses the AI research tools, and creates/manages research projects.
2. **Team Member:** A collaborator invited into a research project - shares tasks, papers, and discussion within that project's workspace.
3. **Supervisor:** Reviews a project's progress, leaves feedback on submitted work, and approves or requests revisions on milestones.
4. **Admin:** Platform manager - monitors overall usage, manages subscription tiers, and oversees AI request quotas and costs across users.

Functional Requirements

SL	Common Workflows
1	Registration and Login System: Users register by providing name, email, and password, and select a role (Researcher or Team Member); Supervisors are invited into a specific project by an existing member rather than self-registering. Once registered, users log in securely via JWT-based authentication to reach their personalized dashboard.
2	Admin Role & Access Management: An authorized Admin signs into a dedicated panel to monitor platform-wide activity - total users, AI request volume, and active subscriptions. The Admin can adjust subscription tier limits, manually resolve payment disputes, and review flagged or unusually high AI usage.

Module 1	
Member	Feature Description
1	Dashboard Overview — Users see a summary card showing total papers uploaded, total projects joined, number of completed papers, and their current subscription plan (Free/Premium).
2	Paper Upload & Library — Researchers upload PDFs to a personal library. Papers can be filtered and organized by topic, year, author, or research area. Custom tags can be added to any paper.
3	Paper Reading Progress Tracker — Tracks each paper's reading status (Unread / Reading / Completed), read percentage (0–100%), and total time spent reading. Visible from the library list view.
4	Research Project Creation — Researchers create named projects (e.g. "Federated Learning for Pest Classification") with a title and description. The creator becomes the project owner automatically.

Module 2	
Member	Feature Description
1	Research Journal — A private, dated log where researchers record daily research progress, notes from meetings, and updates on papers and projects. Entries can be tagged with a paper title using autocomplete from the user's library, linking that log entry directly to the relevant paper.
2	Paper Overview + AI Chat with Paper — Displays an abstract panel and page-anchored notes for a selected paper, alongside a Gemini-powered Q&A chat scoped specifically to that paper's content.

3	Discussion Board + Project Membership & Invitations — Project owners invite collaborators or supervisors by email and assign their role, with unaccepted invites automatically marked “Expired” after 7 days and available for one-click resend. Once a member joins, they gain access to a per-project discussion board with timestamped, attributed threaded posts, and @mentions trigger a notification badge on the mentioned member’s dashboard.
4	AI Paper Summarization + AI Title Generator - Generates a 3-section summary (abstract/methodology/findings) and separate bullet lists for contributions, advantages, limitations, and future work using the Gemini API. Given a topic or short project description, the system generates five candidate paper or thesis titles via the Gemini API. Each generated item includes a save/star icon, so favorites persist to the project instead of being lost when the user navigates away.

Module 3	
Member	Feature Description
1	Bookmarking + Related Paper Recommendations — Users can bookmark papers into a dedicated saved-papers view, and the system surfaces a “More like this” panel based on tag and title overlap with the user’s library. Users can dismiss a bad recommendation, and dismissed papers are permanently excluded from future suggestions for that user.
2	Task Management + Research Timeline Tracker — A per-project task board lets members create, assign, and update tasks, which automatically feed into a visual milestone timeline (Proposal, Mid-Defense, Final Paper,

	Defense). Any task that passes its due date without being marked Done is visually flagged as overdue on both the board and the timeline.
3	Research Gap Finder — Synthesizes the limitations and future-work sections across two or more user-selected papers to generate candidate research gaps using the Gemini API. Each generated gap displays a small source tag indicating which of the selected papers it was derived from, building on the Key Insight Extraction data from Module 2.
4	Key Insight Extraction + Research Question Generator - A “Regenerate” button lets users re-run the summary if the first pass misses something, keeping only the latest version, while each insight bullet has an edit icon so users can correct or shorten a point without regenerating the entire list. Given a topic or short project description, the system generates candidate research questions/hypotheses via the Gemini API. Each generated item includes a save/star icon, so favorites persist to the project instead of being lost when the user navigates away.

Module 4	
Member	Feature Description
1	Citation Generator — Generates formatted citations (APA, IEEE, or ACM style) for any paper in the user’s library, ready to copy or export. This removes the need for users to manually format references when writing papers or reports.
2	Literature Comparison Matrix — Auto-compiles a comparison table (Title, Dataset, Method, Result) from a set of user-selected papers using the Gemini API, drawing on data already extracted during AI

	Summarization and Insight Extraction. The table supports one-click export to CSV for use outside the platform.
3	Supervisor Milestone Approval + Meeting Scheduler — Project members request meeting time slots for supervisor consultations, which the supervisor can accept or propose an alternative for; supervisors can also review submitted milestones and approve them or request revisions. An approved milestone automatically updates its status on the Module 3 research timeline, so members don't need to update it in two places.
4	Notification Center — Aggregates all system-generated signals into a single view, including overdue-task flags, @mention badges, expired-invite alerts, and milestone approval updates. Since it only reads and consolidates data already produced by earlier modules, it depends on Modules 2 and 3 without requiring any new AI or backend logic of its own.

Additional Features

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